Lab 6: ARIMA

MATH840 | Time Series | Week 7 — graded assignment 5 of 7

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This week: accuracy is measured, not marked

Your forecast is scored on the hidden horizon as usual and the score is published the next morning, but **it carries no points this week**. The three points that went to accuracy in Lab 5 go to the modelling itself. Why, in numbers, is below. Accuracy points return in Lab 7.

Open the notebook

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

The practice session's four worked examples, with the same helpers: Open in Colab

The task

Forecast **your series from Week 4** with ARIMA, as the *ARIMA Models, Part 1* lecture defines it:

- differencing, including **one seasonal difference** if the tests ask for it;
- a **non-seasonal** ARMA(p, q) on the differenced series;
- optionally a log transformation, if the seasonal swing grows with the level;
- **no seasonal AR or MA terms**: $P = Q = 0$. Those are the *Part 2* lecture and Lab 7.

The order matters, and it is marked: **difference, read, specify by hand** — and only then compare with `AutoARIMA`.

No ETS, no regression, no averaging with other methods.

This one counts

8 points. **Deadline: 23:59 today**, on Moodle. Marks and comments come back before the next practice session.

What you are marked on

Section	Criteria	Points
1	EDA and visualisation	1
2	Model and forecast	3

Section	Criteria	Points
3	Justification block	3
4	Code quality and reproducibility	1
	Total	**8**

The notebook has one section per row. **Keep those headings.**

> ⓘ **Why accuracy is not marked this week — and is again in Lab 7**
>
> Every series in the pool was issued only if a **seasonal** `AutoARIMA` beats its benchmark. This week's models have no seasonal AR or MA terms, so that guarantee does not cover them. Measured on the whole pool, with exactly this lab's rules:
>
Result against your benchmark	Share of the pool	Points it would earn
beat it	58–60%	3
within 10%	24–26%	2
worse	16%	1
full seasonal `AutoARIMA`, for comparison	100% beat it	3
>
> Whether you lose a point here depends mostly on **how strong the seasonal autocorrelation in your series is** — the practice session's gas example — not on how well you work. So this week the points go to the differencing, the specification and the argument. The score is still published: it is the number Lab 7, with seasonal terms, will be compared against.

📊 Section 1 — EDA and visualisation (1 point)

You know this series by now. The plots this week have one job: to justify the **differencing** and the **transformation** before any model is fitted.

- A time plot, read for trend, season, and whether the seasonal swing grows with the level. If it grows, that is your case for `log=True` — ARIMA has no multiplicative season, and the log is how it gets one.
- The validation split on a plot: history, validation window, your chosen model's forecast and 80% interval over it.
- One sentence per plot. **A plot with no reading attached earns nothing.**

📐 Section 2 — Model and forecast (3 points)

Item	Points	What earns them
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2.1	1	**Differencing.** `difference_plan` with `M` from the setup cell, on y or $\log y$ as Section 1 argued; the differenced series plotted. Overriding a test is allowed and must be argued in 3.1.
2.2	1	**Specification and comparison.** ACF and PACF of the differenced series out to lag $3m$; **at least two hand-specified candidates** written down from that reading; `AutoARIMA` added; all compared with `compare` and decided with `rolling_mase`, the benchmark in the same table.
2.3	1	**The submission.** The chosen candidate refitted on the **whole** history, h steps, with the model's 80% interval, written to the file by the notebook.

⚠ **Where this goes wrong**

- **Fitting outside `fit_arima`.** It is the only builder in the template, and it refuses seasonal AR and MA terms. `AutoARIMA(season_length=M)` on its own will add them — that model is Lab 7's.
- **Comparing AICc across different d, D or log choices.** AICc is comparable only between models fitted to the same differenced data. Across those choices, `rolling_mase` decides.
- **Ljung-Box with `model_df=0`.** The template's `compare` uses $p+q$ — use the same if you test by hand.
- **Reading the residual spike at lag m as a mistake.** With $P=Q=0$ it is often expected: the seasonal difference over-corrects. Report it in 3.3; Lab 7 is where it gets fixed.
- **Forecasting from `train`.** The submission is refitted on the whole history.

📖 FPP3 §9.1–9.8 · FPP3 §5.10 cross-validation · FPP the Pythonic Way, ch. 9

✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

Item	Points	What earns them
3.1	1	**The differencing.** D, d and the log decision, each tied to a test or a plot. If you overrode a test, why — and what the ACF of the result looked like.
3.2	1	**The hand reading and the machine.** The lags in the ACF and PACF that named your candidates; what `AutoARIMA` chose and why it agrees or disagrees with you; the candidate you submitted and the rolling-origin evidence for it.
3.3	1	**What is left, and what you expect.** Ljung-Box with $p+q$ degrees of freedom, read; the residual autocorrelation at lag m and what it says about $P=Q=0$; your expected skill against the benchmark, and how this model compares with your Lab 5 ETS on the validation window.

⚠ **Important**

Marked on the reasoning. A hand specification that `AutoARIMA` beats, with a clear account of why, scores full marks. So does a series where the honest answer is a random walk — if the ACF of the differences is what you show for it.

🖌️ Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: candidates are compared on `train` / `valid` and `rolling_mase`; only the final model is fitted on all of `y`.
- Every number in your text comes from a cell.

📤 Submission

Upload three files to the Week 7 activity on Moodle **before 23:59 today**:

- `SURNAME_lab07_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
`student_id`	your identifier
`code`	your series code, the same as in Weeks 4 and 6
`ds`	the dates you are forecasting, in order
`yhat`	your point forecast
`yhat_lo_80`, `yhat_hi_80`	your model's 80% interval

- `SURNAME_lab07.pdf` — the report, with the justification block;
- `SURNAME_lab07.ipynb` — the source.

> ⚠️ **The format matters even though accuracy does not**
>
> Item 2.3 is the submission file: exactly h rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine.

⏱️ How long this should take

Ninety minutes if you follow the practice session's examples. `rolling_mase` refits each candidate three times; with four candidates on a monthly series that is under a minute.

For next week

Lab 7 is seasonal ARIMA, on this same series — the *ARIMA Models, Part 2* lecture, already published. Accuracy is marked again, and every series in the pool is one a seasonal ARIMA is known to beat. Before then: look at the residual ACF of today's model at lags m, $2m$, $3m$. Whatever is there is what P and Q are for.